This article has been accepted for publication in IEEE Geoscience and Remote Sensing Letters. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/LGRS.2024.3373038

Global-margin Uncertainty and Collaborative Sampling for Active Learning in Complex Aerial Images Object Detection

Dongjun Zhu, Chengjie Gu, Junjun Zhang, Yuyou Yao, Dayu Tan

***Abstract*—Object detection in aerial images based on deep learning requires a large amount of labeled data, whereas manual annotation of aerial images is time-consuming and laborious. As a branch of machine learning, active learning can help humans find the valuable samples by designing some corresponding query strategies, which effectively reduces the cost of manual labeling. However, objects in aerial images are usually small, dense, and accompanied by the interference from complex backgrounds. These brings considerable challenges for active learning in selecting high-value aerial image samples. Currently, there is a relatively lack of study on active learning for aerial images object detection. Therefore, this paper proposes an novel active learning method, using global-margin uncertainty (GMU) and collaborative sampling (CS) to find out the high valuable aerial image samples to reduce the annotation cost and improve the training efficiency of models. In GMU, the predicted scores of categories are applied to calculate the global uncertainty and margin uncertainty of unlabeled aerial images, then those aerial images with high uncertainty scores are selected as the candidate samples. In CS, we train a main model and an auxiliary model respectively to detect the candidate samples, where the samples with large differences in detection results of the two models are selected for manual annotation. The experiments conduct on VisDrone2019 and DOTA-v1.5 datasets, which showes that the proposed method has a better performance compared with several state-of-the-art active learning methods.**

***Index Terms*—Aerial images, object detection, active learning, global-margin uncertainty, collaborative sampling.**

I. Introduction

Object detection is used to identify and locate the interest objects in images. With the development of deep learning, object detection based on deep learning [1], such as RC-NN series [2], YOLO series [3], and outstanding Transformer series [4], shows an excellent performance on remote sensing analysis. Xue et al. [4] propose a dual network structure with interweaved global-local feature hierarchy based on the transformer architecture (DIAG-TR), to alleviate the incompatibility of global and local feature form, and hierarchically embed the local features into global representations. The experiments proves that the proposed method has great potential in earth observation community. To address the common inconsistency between the classification score and localization accuracy, Han et al. [5] propose a single-shot alignment network (S2A-Net) consisting of two modules: a feature alignment module (FAM) and an oriented detection module (ODM), and achieve the state-of-the-art performance on two commonly used aerial objects data sets while keeping high efficiency. To circumvent the two issues, 1) the notorious boundary discontinuity problem, 2) the inconsistency in regression schemes between the two stages, Yao et al. [6] propose a simple and effective bounding box representation by drawing inspiration from the polar coordinate system and integrate it into two detection stages, and obtain the best results in the experiments. Cheng et al. [7] presents a two-stage oriented object detection method, termed dual-aligned oriented detector (DODet), toward evading the aforementioned problems of spatial and feature misalignments. These above methods have made significant contributions to in remote sensing images object detection.

However, training deep learning model requires a large amount of labeled data. Compared with natural images [8] [9], the aerial images usually include smaller and dense objects in a large complex background. This makes the manual labeling much laborious. Fig. 1 shows the comparison of annotation information between natural image datesets (VOC2012 [8], COCO2017 [9]) and aerial image datasets (VisDrone2019 [10], DOTA-v1.5 [11]), where “relative size of object” means the ratio of object area to background area.

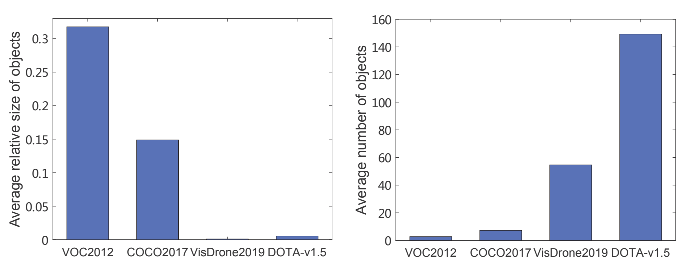

Fig. 1. Average number and relative size of objects in images of different datasets.

This work was supported in part by the Scientific Research Foundation for High-level Talents of Anhui University of Science and Technology 2023yjrc33, and in part by the Natural Science Research Project of Colleges and Universities in Anhui Province 2023AH051197 (Corresponding author: Chengjie Gu.)

Dongjun Zhu, Junjun Zhang, and Yuyou Yao are with School of Public Security and Emergency Management, Anhui University of Science and Technology, Huainan, 232001, China (e-mail: dongjun.zhu@aust.edu.cn, junjun.zhang@aust.edu.cn, yaoyy@aust.edu.cn).

Chengjie Gu is with School of Public Security and Emergency Management, Anhui University of Science and Technology, Huainan, 232001, China, and also with the School of Information Science and Technology, University of Science and Technology of China, Hefei, 230027, China (e-mail: gcj@ustc.edu.cn).

Dayu Tan is with the Key Laboratory of Intelligent Computing and Signal Processing, Ministry of Education, Institutes of Physical Science and Information Technology, Anhui University, Hefei, 230601, China (e-mail: tandayu19@163.com).

This article has been accepted for publication in IEEE Geoscience and Remote Sensing Letters. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/LGRS.2024.3373038

In Fig. 1 (a), the average objects' relative size in aerial images datasets is smaller than that in natural images datesets. Especially in Visdrone2019, the average relative size of objects is 0.0024 which is 130 times smaller than that (0.31789) in VOC2012. On the other side, the Visdrone2019 and DOTA-v1.5 have much more objects than VOC2012 and COCO2017, as shown in Fig. 1 (b). Therefore, selecting and annotating these redundant remote sensing data manually is time-consuming and laborious. It is significant to explore some effective methods to reduce the labor cost of aerial images annotation, and utilize a small number of high-quality samples to obtain a high-performance model.

As a branch of machine learning, active learning [12] aims to design some effective query strategies to select the high-value samples which are beneficial to model performance from the unlabeled data pool. Then those selected samples are labeled by human and added into the original dataset to train model iteratively. Since active learning can assist humans to find high-value samples, effectively reducing the labor cost and improving the model training efficiency. Therefore, active learning has gradually received the researchers' attention in aerial image object detection.

Currently, the main bottlenecks of active learning methods is that the prediction information can not be fully and effectively used to evaluate the value of unlabeled samples. Active learning for object detection usually leverages the entropy or predicted scores as the uncertainty of objects [13] to evaluate the samples' value. Qu et al, [14] leverage the class-imbalance and difference of objects in each images to estimate the uncertainty of unlabeled samples. Lv et al. [15] proposed a active learning method for defect detection, this method leverages uncertainty sampling to generate the candidate list of unlabeled samples, and designs an average margin method to set the sampling rate for each defect category. Desai et al. [16] designed a fine-grained sampling approach for active learning in object detection, which aims to pick the most informative subset of bounding boxes selectively, rather than a entire image. Those above active learning methods show good performance on object detection. However, the global and margin relation in classification uncertainty is not considered sufficiently.

Furthermore, most active learning methods only consider the classification uncertainty in object detection [17], ignore the uncertainty of position regression. Some researchers try to use the two-stage object detection models [2] to find the position difference between the region proposed and the final prediction [18]. However, these methods are not suitable for one-stage object detection models. Some researchers try to integrate active learning into the deep learning models [19] and design some corresponding network structure [20] to predict the uncertainty of samples. Liu et al. [21] proposed a deep reinforcement active learning method, utilizing an agent to select the training samples and taking the reward of reinforcement learning as the uncertainty value of unlabeled samples. Yuan et al. [22] proposed a multiple instance active object detection, which trains two adversarial instance classifiers on the labeled dataset and leverages the discrepancy of them to predict instance uncertainty in the unlabeled dataset, then re-weight instances to estimates the image uncertainty.

Although this method can get good results, there is a high coupling between the original models and active learning methods, which is unfriendly when models are changed.

In this paper, we propose a novel active learning method for aerial image object detection. This proposed method includes two part: global-margin uncertainty (GMU) and collaborative sampling (CS). In GMU, we first use the entropy of objects' category scores to estimate the global uncertainty of predicted objects. Then the top three category scores of predicted objects are used to analyze the margin uncertainty. We consider the predicted objects, which top1 and top2 scores are close while top2 and top3 scores get a large gap, have high marginal uncertainty. Therefore, we combine the global uncertainty and margin uncertainty to estimate the classification uncertainty of each unlabeled image. After that, we sort these images in descending order according to GMU and select top m as the candidate samples. In CS, we train a main object detection model (abbreviated as MODM) and an auxiliary object detection model (abbreviated as AODM) to detect these candidate samples, and utilize the difference of objects' location and category in two prediction results to estimate uncertainty of candidate samples further. In each round of training, we measure the difference of two prediction results by setting a threshold, and select top n samples with the largest difference for labeling. After that these selected samples are added into the datasets for the next iteration. The experiments are carried out on two aerial image object detection datasets (Visdrone2019 and DOTA-v1.5) to verify the effectiveness of the proposed method.

The contributions of the proposed method are summarized as follows:

1) We propose the GMU, combining both global uncertainty and margin uncertainty of the predicted objects to estimate the uncertainty of unlabeled images.

2) We propose the CS, using the difference of two predicted results of the MODM and AODM to estimate uncertainty of unlabeled images from the candidate sample set.

3) A kind of hierarchical sample query strategy (GMU-CS) is proposed, firstly using GMU to select candidate samples from the unlabeled data pool, then applied CS to query the final samples based on the candidate samples.

The rest of this paper is structured as follows. The related works about active learning and object detection are reviewed in Section 2. The details of the proposed method are described in Section 3. Section 4 is the experimental results and analysis. Section 5 is the conclusion of this work.

II. Proposed Method

In this section, we introduce the proposed method in detail. The overall framework of the proposed method is shown in Fig. 2. It is mainly divided into two parts: global-margin uncertainty (GMU) and collaborative sampling (CS), the detail is as follows.

A. Global-margin Uncertainty

To find out high-value samples from unlabeled data, some methods use entropy, loss values or least confidence to e-

This article has been accepted for publication in IEEE Geoscience and Remote Sensing Letters. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/LGRS.2024.3373038

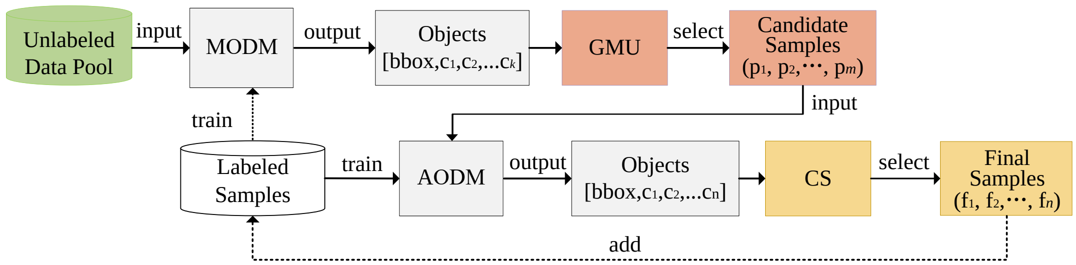

Fig. 2. The framework of GMU-CS.

valuate the uncertainty of unlabeled data. However, these methods didn't utilize classification and regression results effectively. There is a bias in the uncertainty assessment of samples. Therefore, we propose GMU, combining the global and marginal information of the classification results to assess the uncertainty of unlabeled data.

In GMU, we use entropy to measure the global uncertainty of predicted objects, then apply the gap of marginal scores to further mine the value of margin part of predicted objects. The process is shown in Eq. (1), Eq. (2) and Eq. (3).

$$E(O_i) = -\sum_{j=1}^{c} p_j^i \cdot log(p_j^i) \tag{1}$$

In Eq. (1), O_i represents i-th predicted object in an aerial image. p_j^i is the score of category j of O_i, c is the number of categories. $E(O_i)$ denotes the entropy of the object O_i.

We use the top 3 highest scores among all categories to calculate the margin uncertainty of the predicted objects. Let s_1, s_2 and s_3 represent the top 3 category scores of the predicted object. We think that the smaller $(s_1 - s_2)/(s_2 - s_3)$, the higher uncertainty of the margin part. Eq. (2) shows the uncertainty of margin part,

$$M(O_i) = \left(\frac{s_2 - s_3 + \sigma}{s_1 - s_2 + \sigma}\right)^{\mu} \tag{2}$$

where σ represent a constant and is set to 0.001, which is mainly used to avoid the denominator to be zero. μ is a hyperparameter, used to control the weight of margin part in GMU.

Combining the Eq. (1) and Eq. (2), we can estimate the global-margin uncertainty of each predicted object, as shown in Eq. (3),

$$GMU(O_i) = M(O_i) * En(O_i) \tag{3}$$

By summing the GMU of all predicted objects in the unlabeled image img, the GMU of img can be represented as Eq. (4).

$$GMU_{img} = \sum_{i \in I} GMU(O_i) \tag{4}$$

where I represents the set of predicted objects in the unlabeled image img. Finally, we sort the images from large to small according the value of GMU_{img}, and chooses top m images as the candidate samples for the next query.

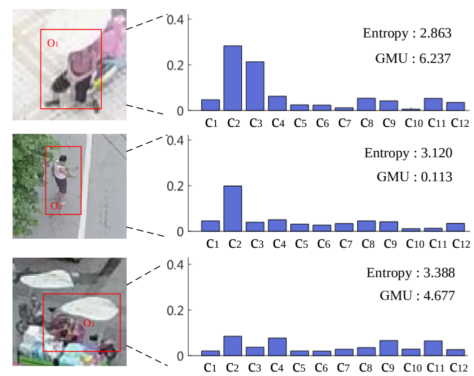

Fig. 3. Global-margin Uncertainty.

Fig. 3 shows the comparison of GMU and entropy, where the abscissa represents the categories, and c_1 -c_{12} corresponds to ignored regions, pedestrian, person, car, van, bus, truck, motor, bicycle, awning-tricycle, tricycle, and others respectively. The ordinate is the category scores of the predicted object.

In Fig. 3, the entropy of predicted objects (O_1, O_2, O_3) are 2.8, 3.1, 3.3, respectively, where the larger of entropy, the more uncertainty of predicted objects are. According to the ranking of entropy values, O_3 is considered the most valuable in these predicted objects.

In O_1, the scores of c_1 and c_2 are very close and much higher than other categories' scores. Therefore, O_1 has the higher uncertainty in margin part (c_1 and c_2). Although the entropy of O_2 is high than O_1, O_2 is easier to be identified due to c_2's score is significantly prominent compared with other categories' scores. For this reason, we think that O_1 is more valuable for the model training. Sorting large to small according to GMU values, the ranking is O_1(6.237), O_3(4.677), O_2(0.113). Therefore, GMU can better evaluate the uncertainty of the predicted objects.

B. Collaborative Sampling

Object detection includes two tasks: classification and regression. It is difficult to use the regression results to estimate the regression uncertainty without extra coordinate information. While the uncertainty of unlabeled image calculated through the accumulation of the uncertainty of the predict objects directly, are easily disturbed by the number of predicted objects.

Therefore, we design CS query strategy, utilizing the MODM and AODM to detect the candidate samples separately. Then the discrepancy of predicted results of the two models can be used to evaluate the uncertainty of samples both in the classification and regression sides. This can be simply regarded as "Checking the answers in pairs". The process of CS are shown in Eq. (5)-Eq. (6).

Eq. (5) shows the Intersection-over-Union (IoU) of two bounding boxes, which can quantify the degree of overlap between two bounding boxes.

$$\mathrm{IoU}_{b_i,b_j^a} = \frac{area(b_i) \cap area(b_j^a)}{area(b_i) \cup area(b_j^a)} \tag{5}$$

In Eq. (5), b_i represents the bounding boxes detected by the main object detection model. b_i^a represents the bounding boxes predicted by the auxiliary object detection model. The *area*(b_i) is the area of bounding box b_i.

According to the IoU and bounding boxes' classification predicted by two object models, we can estimate uncertainty of unlabeled samples. The process is shown as Eq. (6) and Eq. (7).

$$B_s = \{b_i |\ b_i \in B, \mathrm{IoU}_{b_i,b_j^a} > 0.5\ and\ C(b_i) = C(b_j^a)\} \tag{6}$$

In Eq. (6), $C(b_i)$ denotes the predicted category of b_i. Similarly, $C(b_j^a)$ is the predicted category of b_j^a.

$$B_{img}^d = (B \cup B^a) - B_s \tag{7}$$

In Eq. (6), B represents the set of bounding boxes predicted by MODM, B^a is the set of bounding boxes detected by AODM. When IoU(b_i, b_j^a) is over 0.5, b_i and b_j^a are regarded as the trusted bounding boxes, which means they are predicted correctly in a high probability. Therefore, B_{img}^d represents the set of predicted bounding boxes of image img, which have a high probability to be predicted incorrectly. If B_{img}^d is large, it means that the uncertainty of img is high, and should be selected preferentially.

Combining GMU and CS, GMU-CS is proposed. We first select top m unlabeled images as candidate samples base on GMU, then use CS to select the top n images from candidate samples as the most valuable samples for manual annotation.

III. EXPERIMENTS

In this section, we firstly introduce the datasets and evaluation criteria, then describe the details of experimental setting. After that, we present and analyze the results of ablation experiments. Finally, several commonly state-of-the-art active learning methods are added into the comparative experiments to prove the advanced of the proposed method.

A. Datasets and evaluation

In the experiments, we adopt two widely used object detection datasets of aerial images (VisDrone2019 and DOTA-1.5) to verify the effectiveness of proposed method. VOC2007 11 point metric [8] is used to evaluate the performance of object detection models. Under training with the same samples, the better the performance of models, the more effective the active learning method is. The main details of two datasets are shown in Table I.

TABLE I
DETAILS OF THE TWO DATASETS

Datasets	Train images	Test images	Categories	Resolution
VisDrone2019	6471	548	10	500∼2000
DOTA-1.5	7208	2235	15	800∼1280

B. Implementation Details

In the experiments, we use FCOS [23] as MODM and FoveaBox [24] as AODM, where the backbone of two object detection models is ResNet50 [25] and the learning rate is set as 0.0025. Before the experiments, we randomly select 5% samples from the training images of datasets as the initial labeled sample set, and the rest (95% training images) as unlabeled data pool. In each round, 5% training sets (n samples) are selected from the unlabeled data pool, and added into the labeled sample set. Each experiment was conducted in 9 rounds, with a total of 45% training sets are selected and added to the labeled sample set. Then the object detection models are trained 12 epochs, using the updated sample set.

The experiments are conducted on a graphics workstation (Intel Xeon Gold 5117 CPU, 24 GB memory, and 2 Nvidia 2080ti GPU). The proposed method is implemented based on python and PyTorch. We compare the proposed method with six common methods: random (R), classification uncertainty (C), entropy (E), MI-AOD [22], Core-Set [26], CDAL [27] to prove the effectiveness of the proposed method. The final experiment results are the average of the three times results. The results in the tables and figures are converted to percentage(%), where the best results are shown in bold.

C. Experimental results and analysis

To study the effect of parameter μ on GMU, μ is set to 0.001, 0.1, 0.3, 0.5, 0.7, 1, and 2, respectively. The experiments are carried out on the VisDrone2019 dataset. In each round, we select the samples according to the ranking of GMU values and add these samples into the labeled sample set. Then we use the updated sample set to train MODM and observe the effect of μ. Table II shows the training results under different μ values.

TABLE II
THE PERFORMANCE OF GMU WITH DIFFERENT μ VALUES IN VISDRONE2019.

	Proportion of training samples actually used to the total samples								
μ	10%	15%	20%	25%	30%	35%	40%	45%	50%
0.01	18.02	**20.41**	21.02	21.96	22.76	23.52	23.56	24.47	24.63
0.1	18.24	20.03	20.73	21.63	22.52	22.65	24.36	24.60	25.07
0.3	18.46	20.12	**21.63**	**22.75**	23.23	**24.18**	**24.52**	**24.91**	**25.50**
0.5	25.50	20.05	21.30	22.23	23.07	23.50	24.33	24.63	25.15
0.7	**18.83**	19.50	20.60	21.32	22.97	23.20	23.53	24.61	24.85
1	18.27	20.19	20.77	21.42	**23.52**	23.26	24.13	24.50	25.09
2	17.86	19.59	19.99	21.54	22.30	23.06	23.79	24.19	24.52

As shown in Table II, when μ is 0.3, GMU shows a better performance compared with μ equal to other values.

This article has been accepted for publication in IEEE Geoscience and Remote Sensing Letters. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/LGRS.2024.3373038

In addition, it can be seen that the performance of GMU does not keep well in the last few rounds when μ is 0.001 or 2. The case maybe that the weight of the margin part is too large when μ is 2, oppositely the weight would become too small while μ is 0.01. This makes GMU incline to the global or margin part excessively, lacking a comprehensive assessment of the predicted objects. In this case, the selected samples in last rounds don't have high value for model training, which reduces the effect of model training.

To study the influence of m on CS, m is set as $1.5n$, $2n$, $2.5n$, $3n$ respectively, where m represents the number of samples selected by GMU, n is the number of samples selected by GMU-CS. The experiments are carried out on VisDrone2019 dataset.

TABLE III
THE PERFORMANCE OF GMU-CS WITH DIFFERENT m IN VISDRONE2019.

	Proportion of training samples actually used to all samples (%)								
m	10%	15%	20%	25%	30%	35%	40%	45%	50%
$1.5n$	18.36	20.15	21.51	23.05	22.67	23.78	24.37	24.53	25.62
$2n$	**18.81**	**20.63**	21.65	**23.11**	**23.95**	**24.43**	24.37	**25.59**	**25.93**
$2.5n$	18.73	20.39	**21.77**	23.11	23.67	24.13	**24.98**	25.33	25.71
$3n$	18.57	19.97	21.13	22.67	23.37	24.13	24.53	25.03	25.22

As shown in Table III, GMU-CS exhibits the higher performance in most rounds when m=$2n$. This indicates that when m=$2n$, GMU-CS can utilize the predicted results (classification and position) of two object detection models effectively to mine the high-value samples from the candidate sample set. When m is $3n$, the mAP is gradually reduced in the last several rounds. The case is that some samples with low uncertainty for MODM and high uncertainty for AODM may also be selected, when m becoming too large. These samples are low-value for the model training. Therefore, according to Table III, we use m equal to $2n$ for the subsequent experiments.

TABLE IV
PERFORMANCE COMPARISON OF E, M, GMU AND GMU-CS

	Proportion of Labeled Images (%)								
models	10	15	20	25	30	35	40	45	50
E	17.36	20.03	20.98	21.53	22.79	23.51	23.75	24.51	24.76
M	18.02	20.17	21.26	22.02	22.27	23.67	24.13	24.55	24.93
GMU	18.46	20.12	21.63	22.75	23.23	24.18	24.52	24.91	25.50
GMU-CS	**18.81**	**20.63**	**21.65**	**23.11**	**23.95**	**24.43**	**24.91**	**25.59**	**25.93**

Table IV shows the performance comparison among E, M, GMU, and GMU-CS, where E represents entropy method used as the baseline, M represents the query strategy based on Eq. (2). As shown in Table IV, GMU can achieve better performance compared with E in most rounds, which indicates that combining the global and margin information to select samples can mine the high-value samples efficiently. In Table IV, GMU-CS shows a better detection accuracy compared with CS and GMU. GMU-CS not only uses the classification information but also utilizes the position information of objects predicted by two object detection models. This makes GMU-CS find the high-value of samples for model training more accurately.

As shown in Fig. 4 and Fig. 5, the dotted line represents the accuracy of MODM trained by the full training set. From

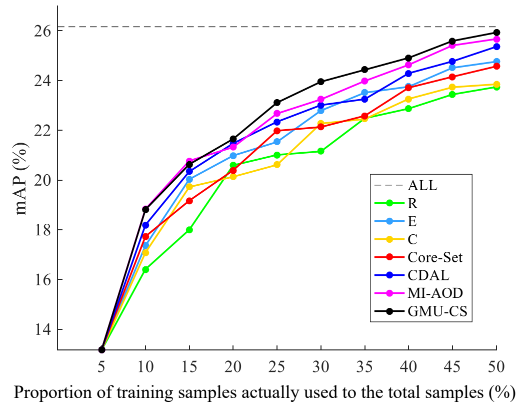

Fig. 4. Comparison of the detection results with different active learning methods on Visdrone2019 dataset

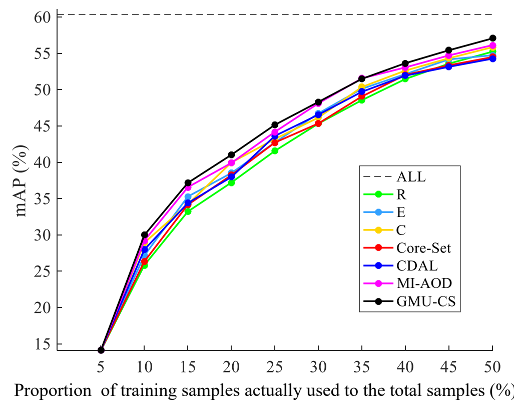

Fig. 5. Comparison of the detection results with different active learning methods on DOTA dataset

the experimental results on the two data sets, GMU-CS can make the object detection models achieve higher detection accuracy compared with other active learning methods. As shown in Fig. 4, GMU-CS only uses 50% training samples of Visdrone2019 and makes mAP of MODM almost close to that performance trained with all training samples. In DOTA v1.5, as shown in Fig. 5, GMU-CS achieves 96% performance that trained by all training samples while only using 50% of all training samples. This indicates that GMU-CS has the good robustness on different datasets. GMU-CS combines global and marginal classification information, and add a AODM to collaboratively sample the uncertainty samples. This makes GMU-CS can use prediction information of classification and regression effectively, and find out those samples which more valuable for model training.

Fig. 6 shows the example of visualized detection results using different active learning methods on Visdrone2019 dataset. As shown in Fig. 6, the object detection model trained on the samples set selected by GMU-CS can achieve the almost similar detection performance compared with the model trained the full training set. Comparatively, while using the other active learning methods, the trained object detection model is prone to losing more objects. This further demonstrates the advantage of GMU-CS in selecting high-quality samples.

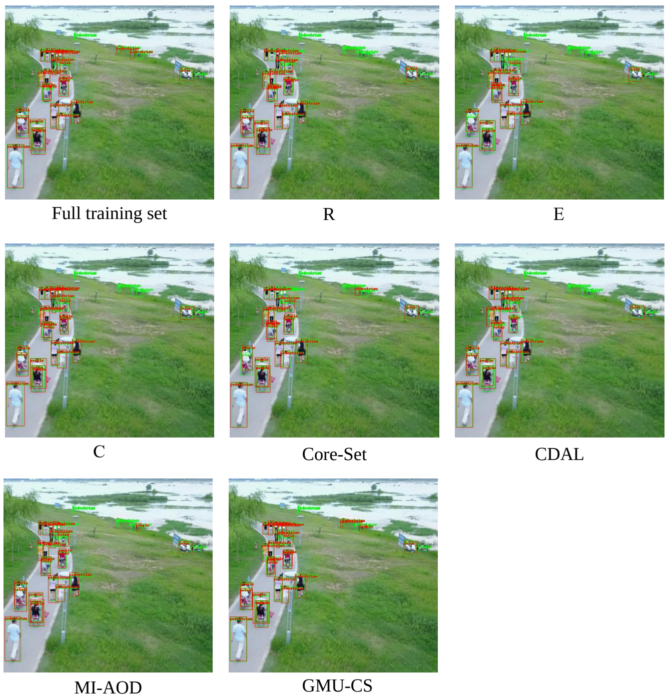

Fig. 6. Visualized detection results using different active learning methods on Visdrone2019 dataset.

IV. Conclusion

In this paper, we propose a novel active learning method GMU-CS for aerial images object detection, where GMU utilizes the global and marginal information of classification prediction to estimate uncertainty of unlabeled samples, and select top m samples from the unlabeled data pool as the candidate sample set. To use classification and regression information effectively, CS are proposed, using the difference of detection results between MODM and AODM to select high-value samples from the candidate sample set further. Finally, these selected samples are labeled and put into the labeled dataset for the modeling training. The experiments based on VisDrone2019 and DOTA-v1.5 show that GMU-CS has better performance compared with several state-of-the-art active learning methods, and can approximate the training effect of the full training set, only using half of the training set. In future work, we try to apply active learning into federated learning to mine more valuable samples for each client to improve the performance of clients' models.

References

[1] X. Wu, D. Sahoo, and S. C. Hoi, "Recent advances in deep learning for object detection," *Neurocomputing*, vol. 396, pp. 39–64, 2020.

[2] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *Advances in neural information processing systems*, vol. 28, 2015.

[3] L. Liu, W. Ouyang, X. Wang, P. Fieguth, J. Chen, X. Liu, and M. Pietikäinen, "Deep learning for generic object detection: A survey," *International journal of computer vision*, vol. 128, no. 2, pp. 261–318, 2020.

[4] J. Xue, D. He, M. Liu, and Q. Shi, "Dual network structure with interweaved global-local feature hierarchy for transformer-based object detection in remote sensing image," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 15, pp. 6856–6866, 2022.

[5] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–11, 2021.

[6] Y. Yao, G. Cheng, G. Wang, S. Li, P. Zhou, X. Xie, and J. Han, "On improving bounding box representations for oriented object detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–11, 2022.

[7] G. Cheng, Y. Yao, S. Li, K. Li, X. Xie, J. Wang, X. Yao, and J. Han, "Dual-aligned oriented detector," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–11, 2022.

[8] M. Everingham, L. Van Gool, C. K. Williams, J. Winn, and A. Zisserman, "The pascal visual object classes (voc) challenge," *International journal of computer vision*, vol. 88, no. 2, pp. 303–338, 2010.

[9] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *European conference on computer vision*. Springer, 2014, pp. 740–755.

[10] P. Zhu, L. Wen, D. Du, X. Bian, Q. Hu, and H. Ling, "Vision meets drones: Past, present and future," *arXiv preprint arXiv:2001.06303*, 2020.

[11] G.-S. Xia, X. Bai, J. Ding, Z. Zhu, S. Belongie, J. Luo, M. Datcu, M. Pelillo, and L. Zhang, "Dota: A large-scale dataset for object detection in aerial images," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 3974–3983.

[12] S. Budd, E. C. Robinson, and B. Kainz, "A survey on active learning and human-in-the-loop deep learning for medical image analysis," *Medical Image Analysis*, vol. 71, p. 102062, 2021.

[13] C. C. Aggarwal, X. Kong, Q. Gu, J. Han, and S. Y. Philip, "Active learning: A survey," in *Data Classification: Algorithms and Applications*. CRC Press, 2014, pp. 571–605.

[14] Z. Qu, J. Du, Y. Cao, Q. Guan, and P. Zhao, "Deep active learning for remote sensing object detection," *arXiv preprint arXiv:2003.08793*, 2020.

[15] X. Lv, F. Duan, J.-J. Jiang, X. Fu, and L. Gan, "Deep active learning for surface defect detection," *Sensors*, vol. 20, no. 6, p. 1650, 2020.

[16] S. V. Desai and V. N. Balasubramanian, "Towards fine-grained sampling for active learning in object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2020, pp. 924–925.

[17] S. Sivaraman and M. M. Trivedi, "Active learning for on-road vehicle detection: A comparative study," *Machine vision and applications*, vol. 25, no. 3, pp. 599–611, 2014.

[18] C.-C. Kao, T.-Y. Lee, P. Sen, and M.-Y. Liu, "Localization-aware active learning for object detection," in *Asian Conference on Computer Vision*. Springer, 2018, pp. 506–522.

[19] G. Hacohen, A. Dekel, and D. Weinshall, "Active learning on a budget: Opposite strategies suit high and low budgets," in *International Conference on Machine Learning*. PMLR, 2022, pp. 8175–8195.

[20] X. Di, Z. Xue, and M. Zhang, "Active learning-driven siamese network for hyperspectral image classification," *Remote Sensing*, vol. 15, no. 3, p. 752, 2023.

[21] Z. Liu, J. Wang, S. Gong, H. Lu, and D. Tao, "Deep reinforcement active learning for human-in-the-loop person re-identification," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 6122–6131.

[22] T. Yuan, F. Wan, M. Fu, J. Liu, S. Xu, X. Ji, and Q. Ye, "Multiple instance active learning for object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 5330–5339.

[23] Z. Tian, C. Shen, H. Chen, and T. He, "Fcos: Fully convolutional one-stage object detection," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 9627–9636.

[24] T. Kong, F. Sun, H. Liu, Y. Jiang, L. Li, and J. Shi, "Foveabox: Beyound anchor-based object detection," *IEEE Transactions on Image Processing*, vol. 29, pp. 7389–7398, 2020.

[25] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[26] O. Sener and S. Savarese, "Active learning for convolutional neural networks: A core-set approach," in *International Conference on Learning Representations*, 2018.

[27] S. Agarwal, H. Arora, S. Anand, and C. Arora, "Contextual diversity for active learning," in *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XVI 16*. Springer, 2020, pp. 137–153.